



ASSET LOADING · SCENE SETUP

Choose a blue sky backdrop and delete the default cat



STEPS TO FOLLOW

1. Open Scratch → right-click cat → Delete
2. Choose a Backdrop → search "Blue Sky"
3. Click to add → stage turns sky blue!

TIP

Any bright sky backdrop works. "Blue Sky 2" has clouds already painted in — saves you one step!

DID YOU KNOW?

Loading assets and configuring the scene before the game loop starts is called scene setup or initialisation — every game engine has this explicit phase.



OBJECT PROPERTIES • POSITIONING

Add a parrot or bird sprite, size it to 50, and position it on the left of the stage



SCRATCH BLOCKS — ON THE BIRD SPRITE

when  clicked

go to x: (-180) y: (0)

TIP

Placing the bird at x=-180 gives it room to navigate. Pipes will come from the right side.

DID YOU KNOW?

Every game object starts with an initialization block that sets position, size, and velocity to known values before the game loop begins.



VARIABLES · PHYSICS SIMULATION

Create a "y velocity" variable; every frame subtract 1 (gravity) and apply it to y position



SCRATCH BLOCKS — ON THE BIRD SPRITE

when clicked

go to x: (-180) y: (0)

set [y velocity] to (0)

forever

change [y velocity] by (-1)

change y by (y velocity)

if <(y position) < (-170)> then

set y to (-170)

set [y velocity] to (0)

TIP

Try "change [y velocity] by (-2)" for stronger gravity — the game gets much harder immediately!

DID YOU KNOW?

Velocity accumulation under constant acceleration is Newtonian mechanics — $F=ma$ in code. Every physics engine from Box2D to Unreal uses this exact loop every frame.



EVENT HANDLING · INPUT POLLING

Press SPACE to set y velocity to +8 — fighting gravity with a flap impulse



SCRATCH BLOCKS — SEPARATE SCRIPT ON BIRD

– *separate script on BIRD* →

when [space] key pressed

set [y velocity] to (8)

TIP

Try values like 6, 10, or 15 to find the feel you want. Higher = easier to stay aloft but harder to aim through gaps.

DID YOU KNOW?

"when [space] key pressed" is a hardware interrupt wrapper — the operating system tells Scratch the instant the key is pressed, with no polling delay.



OBJECT TEMPLATES · HIDING

Draw a green pipe sprite in the Paint Editor and hide it on flag click



SCRATCH BLOCKS — ON THE PIPE SPRITE

— *PIPE*: when  clicked —

hide

TIP

Make the pipe width about 28–32 pixels. Too wide = impossible game. Too thin = too easy.

DID YOU KNOW?

The Pipe sprite is a class. Its clones are instances. OOP in action — one template, infinite objects.



REPEAT-UNTIL · DESTRUCTION

**A pipe clone scrolls from right to left and deletes itself when
offscreen**



SCRATCH BLOCKS — ON THE PIPE SPRITE

— PIPE: when I start as a clone —

show

repeat until $\langle (x \text{ position}) < (-240) \rangle$

change x by (-4)

delete this clone

TIP

Increase the scroll speed to -6 or -8

for a harder game. This one number

controls overall game difficulty!



DID YOU KNOW?

Parallax scrolling, endless runners,

and side-scrollers all use this pattern:

spawn off-screen right → scroll left →

destroy off-screen left → repeat.



RANDOM NUMBERS • PROCEDURAL GENERATION

Make the master Pipe spawn clones at random heights in a forever loop



SCRATCH BLOCKS — ON THE PIPE SPRITE

— PIPE master: when clicked —

hide

forever

go to x: (240) y: (pick random (-60) to (60))

create clone of [myself]

wait (2.5) seconds

TIP

Lower the wait to 1.5 secs for a rapid-fire challenge. Wider pick-random range creates more extreme height differences.

DID YOU KNOW?

Procedural generation — Minecraft biomes, Spelunky levels, No Man's Sky planets — is all "pick random" at different scales.



COLLISION DETECTION · GAME OVER

Detect when the Bird touches a Pipe — stop the game immediately



SCRATCH BLOCKS — INSIDE THE PIPE SCROLL LOOP

(inside the repeat-until scroll loop)

change x by (-4)

if <touching [Bird]?> then

stop [all]



TIP

Before "stop [all]", add "say [Game Over! 🌟] for (1) secs" on the Bird sprite for a dramatic ending.



DID YOU KNOW?

"touching [Bird]?" checks pixel-by-pixel overlap 30 times per second — the same AABB collision test behind every hitbox, trigger zone, and collision mesh in game development.



VARIABLES · SCORE STATE

Add a Score variable that increments each time the bird passes a pipe



SCRATCH BLOCKS — INSIDE THE PIPE CLONE SCROLL LOOP

(inside the pipe clone scroll loop)

```
if <(x position) < (-180)> then
```

```
  change [Score] by (1)
```

```
  wait (0.1) seconds
```

```
if <touching [Bird]?> then
```

```
  stop [all]
```

TIP

Share your game on Scratch — click "Share" at the top and challenge friends to beat your high score! 🏆

DID YOU KNOW?

Threshold-crossing detection (" $x < -180$ ") is used in games, finance (stop-loss triggers), IoT (sensor alarms), and analytics (funnel drop-off).